Jay A. Panchal

AI Engineer

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Summary

AI Engineer with 1.5+ years of software engineering experience, building LLM applications, retrieval-augmented generation (RAG) pipelines, and machine learning models in Python. Teaching Assistant for graduate Machine Learning while completing a Master of Science in Computer Science at Loyola Marymount University.

Education

Loyola Marymount University — Master of Science in Computer Science (MS) *Los Angeles, CA | Expected Dec 2026*
GPA: 3.88 | *Coursework*: Generative AI, Machine Learning, Deep Learning, Database Systems, Algorithms

Gujarat Technological University — Bachelor of Engineering in Computer Engineering (BE) *Ahmedabad, India | May 2023*

Professional Experience

Teaching Assistant, Machine Learning and Deep Learning — Loyola Marymount University *Los Angeles, CA | Aug 2025 - Present*

- Sole Teaching Assistant for the graduate Machine Learning course; grade 5 assignments covering written work on Gradescope and coding work on PostCommit, with clear, consistent feedback
- Previously supported the graduate Deep Learning course

Software Engineer — Aequitas Infotech Pvt. Ltd. *Ahmedabad, India | Jan 2024 - Aug 2024*

- Maintained and improved backend features in Java and MySQL for a practice management platform used by tens of thousands of Chartered Accountants across India
- Built custom features for individual clients and fixed client-reported bugs, often during live support calls
- Gathered requirements in client meetings and trained users on the platform in sessions of up to 300 people

Backend Developer — CodeHabit *Gandhinagar, India | Sep 2023 - Dec 2023*

- Built REST APIs with Node.js, Express, and MongoDB for a hospital management system used by 25+ staff, with JSON Web Token (JWT) authentication and role-based access control

Software Engineer Intern — Praxware Technologies *Ahmedabad, India | Jan 2023 - Aug 2023*

- Built a sales and inventory web application with Python, Django, and MySQL, including REST APIs and create, read, update, and delete (CRUD) features

Projects

GridWatch — Energy Affordability Intelligence System | Python, XGBoost, SHAP, RAG, FAISS, OpenAI API, Streamlit Live Demo

- Trained an XGBoost model on Census Bureau, Energy Information Administration, and NOAA data to predict energy-burden risk across 50 California cities, using SHAP to explain each prediction
- Built a RAG pipeline with a FAISS vector index over 5 government assistance programs, so LLM answers are grounded in cited documents

GoTinerary — Agentic AI Trip Planner | Gemini API, FastAPI, React, Google Places API GitHub

- Built solo in one weekend an AI trip planner where Gemini uses tool calling to query the Google Places API on its own and plan trips with real places
- Built the FastAPI back end and React front end, connected through REST API endpoints

LLM Fine-Tuning | Python, Llama, Supervised Fine-Tuning, LoRA, Tinker API

- Fine-tuned Llama models with supervised fine-tuning (SFT) and Low-Rank Adaptation (LoRA) on a university server through the Tinker API

Pokémon Generator — VAE vs DCGAN | PyTorch, Generative Models

- Trained and compared a Variational Autoencoder (VAE) and a Deep Convolutional GAN (DCGAN) on 2,500 images over 300 epochs, fixing GAN mode collapse with spectral normalization

Technical Skills

- **Languages**: Python, Java, JavaScript, TypeScript, SQL, C++
- **Machine Learning**: PyTorch, Scikit-learn, XGBoost, SHAP, pandas
- **Generative AI**: OpenAI API, Gemini API, Claude API, LangChain, RAG, FAISS, Tool Calling, LoRA Fine-Tuning
- **Back-End and Data**: FastAPI, Django, Node.js, REST APIs, PostgreSQL, MySQL, MongoDB
- **Tools**: Git, GitHub, Streamlit, Postman

Awards

- **LA Hacks 2026 — Best Use of Gemini API (MLH)** | Oct 2026: Won for GoTinerary, built solo in one weekend
- **LMU Datathon — 3rd Place ($1,000)** | Mar 2026: Analyzed the NLSY97 dataset with Python, pandas, and regression